

Yesterday we learned ANN structure...

But how does a neural network improve itself?"

"Today we learn the learning process of Deep Learning."

2. OVERALL TRAINING FLOW

Explain complete pipeline first:

ANN Learning Steps

1. Input data
 2. Generate prediction
 3. Compare with actual output
 4. Calculate error
 5. Update weights
 6. Repeat many times
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Teaching Line

"Neural networks learn by reducing mistakes repeatedly."

3. FORWARD PROPAGATION

Definition

Data moves from:

- input layer
→ hidden layer
→ output layer

to generate prediction.

Simple Meaning

"Forward propagation means making prediction."

4. EXAMPLE

Input:

- study hours
- attendance

Network predicts:

- pass probability

5. 🎨 FORWARD FLOW

Input → Hidden Layer → Output

6. 📖 WHAT HAPPENS INSIDE NEURON?

Each neuron:

1. Receives inputs
2. Multiplies with weights
3. Adds bias
4. Applies activation function
5. Produces output

👉 No heavy formula needed

7. 📖 ACTIVATION FUNCTION

Definition

Activation function decides whether neuron should activate or not.

Simple Meaning

“Activation function adds non-linearity.”

8. ! WHY ACTIVATION FUNCTIONS IMPORTANT?

Without activation:

- network behaves like simple linear model

👉 Deep learning power lost

💡 **Teaching Line**

“Activation functions make neural networks powerful.”

9. 🇮🇹 **TYPES OF ACTIVATION FUNCTIONS**

◆ **SIGMOID**

Range

0 to 1

Use

Probability output

Graph Idea

S-shaped curve

Advantage

Easy probability interpretation

Limitation

Vanishing gradient issue

◆ **RELU (MOST IMPORTANT)**

Full Form

Rectified Linear Unit

Rule

Meaning

- Negative $\rightarrow 0$
 - Positive $\rightarrow$ same value
-

Advantages

- ✓ Fast
- ✓ Efficient
- ✓ Most commonly used

💡 Teaching Line

“ReLU is the default activation in modern deep learning.”

10. 📖 OUTPUT PREDICTION

After forward propagation:

👉 Network gives prediction

Example:

Actual Predicted

1 0.7

11. 📖 LOSS FUNCTION

Definition

Loss function measures prediction error.

Simple Meaning

“Loss tells how wrong the model is.”

12. 🇮🇹 TYPES OF LOSS (Basic Only)

◆ Classification

Binary Crossentropy

◆ Regression

Mean Squared Error

👉 Don't go deep mathematically

13. 📖 BACKPROPAGATION

Definition

Process of updating weights using error.

Simple Meaning

"Backpropagation helps network learn from mistakes."

14. 📖 HOW BACKPROP WORKS

1. Calculate error
 2. Send error backward
 3. Update weights
 4. Reduce future error
-
-

15. 📖 OPTIMIZER

Definition

Optimizer updates weights efficiently.

Simple Meaning

"Optimizer helps model learn faster and better."

16. 📖 COMMON OPTIMIZER

◆ ADAM (MOST IMPORTANT)

Features

- ✓ Fast
- ✓ Stable
- ✓ Most commonly used

💡 Teaching Line

“Adam is the default optimizer in many DL projects.”

17. 📁 COMPLETE TRAINING CYCLE

Input Data
↓
Forward Propagation
↓
Prediction
↓
Loss Calculation
↓
Backpropagation
↓
Weight Update
↓
Repeat

18. 📁 EPOCHS

Definition

One complete pass through entire dataset.

Example

- 10 epochs
→ model learns 10 times
-

💡 Teaching Line

“More epochs = more learning opportunities.”

19. 📁 OVERFITTING IN DL

If epochs too high:

- model memorizes data

Mention briefly only

20. 🎯 INTERVIEW CONCEPTS

Q1: What is forward propagation?

👉 Process of generating predictions.

Q2: What is backpropagation?

👉 Updating weights using error.

Q3: Why activation functions important?

👉 Add non-linearity.

Q4: What is ReLU?

👉 Most common activation function.

Q5: What does optimizer do?

👉 Updates weights to reduce loss.